ECON8037: Financial Economics

Week 7 Lecture - Asset Valuation

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Asset Valuation - Liquidity

In these slides we again develop an asset-pricing model in which financial assets are valued not only as claims to streams of consumption goods but for also for their liquidity. By liquidity, in this context, we mean the degree to which an asset is valued as a medium of exchange at the margin.

We will look at Lagos (2010), and in this case when an asset is held partly for its exchange value, the demand for the asset and its price tend to be higher than if the asset were not held for exchange. Its intrinsic rate of return - which takes into account only the stream of consumption goods that the asset represents - will be lower.

Asset Valuation - Liquidity

We also consider an economy with two assets: an equity share and a one-period government-issued risk-free real bill. In the basic setup, assets differ only in their payoffs, and agents are free to choose which assets to use as means of payment in decentralised trades. In this case, the theory unambiguously predicts that someone testing an agent's Euler equation for the risk-free bill using its intrinsic rate of return would find that, at the margin, this agent can gain from transferring consumption from the future to the present. That is, there would appear to be a risk-free rate puzzle

Asset Valuation - Liquidity

We also analyse version of the economy in which legal or institutional restrictions give bonds an advantage over equity as a medium of exchange. In this case, it is possible to show that there are degrees of these restrictions for which someone testing an agent's Euler equation for the intrinsic excess returns would fins that, at the margin, the agent can gain from disinvesting in bills and investing in equity. There would appear to be an equity premium puzzle

Asset Valuation - Lagos

There is a $[0, 1]$ continuum of agents, time is discrete and the horizon infinite. Each period is divided into two subperiods where different activities take place

There are three nonstorable and perfectly divisible consumption goods at each date: fruit, general goods, and special goods. The only durable commodity in the economy is a set of "Lucas trees". The number of trees is fixed and equal to the number of agents. Trees yield fruit in the second subperiod of every period.

Asset Valuation - Lagos

Each agent wishes to maximise

$$\mathbb{E}\left\{\sum_{t=0}^{\infty}\beta^{t}\left[u\left(q_{t}\right)-e\left(\tilde{q}\right)+v\left(y_{t}\right)+U\left(c_{t}\right)-Ah_{t}\right]\right\}$$

Where $\beta \in (0,1)$, q_t and $\tilde{q}_t$are the quantities of special goods consumed and produced in the decentralised market. y_t denotes consumption of general goods, c_t consumption of fruit, h_t the number of hours worked in the second subperiod. and $\mathbb{E}$ is an expectations operator.

Assume $u\left(0\right)=e\left(0\right)0$, $u'>0$, $e'>0$, $v'>0$, $U'>0$, $u''<0$, $e''\geq 0$, $v''\leq 0$ and $U''<0$

Asset Valuation - Lagos

We begin by considering the case where the equity share is the only asset. Let d_t be the quantity of fruit, or "dividends", produced by each tree in period t, and let a_t be an individual agent's share holdings.

Asset Valuation - Lagos

Let $W(a_t, d_t)$ denote the value function of an agent who enters the centralised market holding a_t shares in a period when dividends are d_t, and let $V(a_t, d_t)$ denote the corresponding value when he enters the decentralised market. These value functions satisfy the following Bellman equation:

$$W(a_t, d_t) = \max_{c_t, y_t, n_t, h_t, a_{t+1}} \{U(c_t) + v(y_t) - Ah_t + \beta \mathbb{E} V(a_{t+1}, d_{t+1})\}$$

Subject to

$$c_t + w_t n_t + \phi_t a_{t+1} = (\phi_t + d_t) a_t + w_t h_t$$

$$0 \leq c_t, \quad y_t = n_t, \quad 0 \leq n_t, \quad 0 \leq h_t \leq \bar{n}, \quad 0 \leq a_{t+1}$$

Asset Valuation - Lagos

The agent chooses consumption of fruit (c_t), of general goods (y_t), how many hours of work to demand (n_t) and supply (h_t), and an end-of-period portfolio (a_{t+1}). Dividends $\{d_t\}$ follow a Markov process with $\mathbb{P}\{d_{t+1} \leq x' | d_t = x\} = F(x', x)$. Fruit is used as numeraire: w_t is the real wage, and ϕ_t is the price of a share (ex-dividend).

Asset Valuation - Lagos

Substituting the constraints that hold with equality,

$$W\left(a_t, d_t\right) = \max_{c_t, n_t, a_{t+1}} \left\{ U\left(c_t\right) + v\left(n_t\right) - \frac{A}{w_t}\left[c_t + w_t n_t + \phi_t a_{t+1}\right] + \lambda_t a_t + \beta \mathbb{E} V\left(a_{t+1}, d_{t+1}\right) \right\}$$

Where $\lambda_t \equiv \frac{A}{w_t}\left(\phi_t + d_t\right)$

Asset Valuation - Lagos

Given the buyer and the seller have share holdings a and $\tilde{a}$ respectively, the terms at which they trade in the decentralised market are $[q(a,\tilde{a}), p(a,\tilde{a})]$, where $q(a,\tilde{a}) \in \mathbb{R}_+$ is the quantity of special good traded, and $p(a,\tilde{a}) \in \mathbb{R}_+$ represents the transfers of assets from the buyer to the seller. The value of an agent who enters the decentralised market with share holdings a in a period when the dividend realisation is d, satisfies

$$V(a,d) = \alpha \int \{u[q(a,\tilde{a})] + W[a - p(a,\tilde{a}), d]\} dG(\tilde{a})$$
$$+ \alpha \int \{-e[q(\tilde{a},a)] + W[a + p(\tilde{a},a), d]\} dG(\tilde{a}) + (1-2\alpha) W(a,d)$$

where G denotes the distribution of share holdings

Asset Valuation - Lagos

Consider a meeting in the decentralised market between a buyer holding a_t and a seller holding $\tilde{a}_t$. The terms of trade (q_t, p_t) are determined by Nash bargaining where the buyer has all the bargaining power:

$$\max_{q_t, p_t \leq a_t} \{u(q_t) + W(a_t - p_t, d_t) - W(a_t, d_t)\}$$

Subject to

$$-e(q_t) + W(\tilde{a}_t + p_t, d_t) \geq W(\tilde{a}_t, d_t)$$

The constraint $p_t \leq a_t$ indicates that the buyer cannot spend more assets than he owns.

Asset Valuation - Lagos

Note that $W(a_t + p_t, d_t) - W(a_t, d_t) = \lambda_t p_t$, so the bargaining problem reduces to:

$$\max_{q_t, p_t \leq a_t} \{u(q_t) - \lambda_t p_t\}$$

Subject to

$$-e(q_t) + \lambda_t p_t \geq 0$$

If $\lambda_t a_t \geq e(q^*)$ then the buyer exchange $p_t = e(q^*)/\lambda_t \leq a_t$ of his shares for the first-best quantity q^*. Else, he gives the seller all his shares, $p_t = a_t$, in exchange for the q_t that solves $e(q_t) = \lambda_t a_t$. Note that the outcome depends only on $\lambda_t a_t$, and in particular, it is independent of the seller's asset holdings

Asset Valuation - Lagos

Hence, the solution to the bargaining problem is

$$q\left(\lambda_t a_t\right) = \begin{cases} q^* & \text{if } \lambda_t a_t \geq e\left(q^*\right) \\ e^{-1}\left(\lambda_t a_t\right) & \text{if } \lambda_t a_t < e\left(q^*\right) \end{cases}$$

Using this bargaining solution and the linearity of $W(a, d)$, the value of search can be written more compactly as

$$V(a, d) = \mathcal{S}(\lambda a) + W(a, d)$$

Where $\mathcal{S}(\lambda a) = \alpha\left\{u\left[q(\lambda a)\right] - e\left[q(\lambda a)\right]\right\}$. Observe that $\mathcal{S}$ is twice differentiable everywhere, with $\mathcal{S}' \geq 0$ and $\mathcal{S}'' \leq 0$ (both inequalities are strict for $\lambda a < e\left(q^*\right)$).

Asset Valuation - Lagos

Having characterised the terms of trade in decentralised exchange, we turn to the agent's individual optimisation problem in the centralise market.

The agent's problem in the second subperiod is summarised by

$$W\left(a_t, d_t\right) = \lambda_t a_t + \max_{c_t, n_t} \left\{ U\left(c_t\right) + v\left(n_t\right) - \left(c_t + w_t n_t\right) \frac{A}{w_t} \right\}$$
$$+ \max_{a_{t+1}} \left\{ \beta \mathbb{E} V\left(a_{t+1}, d_{t+1}\right) - \frac{A}{w_t} \phi_t a_{t+1} \right\}$$

Asset Valuation - Lagos

Given that U and v are strictly concave, the unique solution (c_t, n_t) satisfies

$$U'(c_t) = \frac{A}{w_t}$$

$$v'(n_t) = A$$

The first-order condition for the choice of a_{t+1} is

$$\frac{A}{w_t}\phi_t = \beta \mathbb{E} V_a'(a_{t+1}, d_{t+1})$$

From the DM value function we can show that

$$V_a'(a_{t+1}, d_{t+1}) = \left[1 + \alpha\left(\frac{u'[q(\lambda_{t+1}a_{t+1})]}{e'[q(\lambda_{t+1}a_{t+1})]} - 1\right)\right]\lambda_{t+1}$$

Asset Valuation - Lagos

Note that none of the agent's choices depend on his individual asset holdings. We will consider an equilibrium in which all prices are time-invariant function of the aggregate state: $w_t = w(d_t), \phi_t = \phi(d_t)$, and therefore, $\lambda_t = \lambda(d_t) \equiv \frac{A}{w(d_t)}[\phi(d_t) + d_t]$.

Asset Valuation - Lagos

A *recursive equilibrium* is a collection of individual decision rules $c_t = c(d_t)$, $n_t = n(d_t)$, $a_{t+1} = a(d_t)$, pricing functions $w_t = w(d_t)$ and $\phi_t = \phi(d_t)$, and bilateral terms of trade $q_t = q(d_t)$ and $p_t = p(d_t)$ such that:

(a) The decision rules $c(.)$, $n(.)$, and $a(.)$, solve the agent's problem in the decentralised market, given prices

(b) The terms of trade are determined by Nash bargaining, i.e., $q[\lambda(d_t) a_t]$ and $p(d_t) = \min\left[a_t, \frac{e(q^*)}{\lambda(d_t)}\right]$; and

(c) Prices are such that the centralised market clears, i.e., $c(d_t) = d_t$, and $a(d_t) = 1$.

Asset Valuation - Lagos

In this section, we introduce a government-issued one-period risk-free real bill, which we will refer to as a "bond". Let B_t denote the stock of bonds that are outstanding in period t, to be redeemed before the centralised trading session of period t. (The government sells B_{t+1} in the centralised market at the end of period t). What we call the "government", is essentially summarised by the budget constraint

$$B_t = \phi_t^b B_{t+1} + \tau_t$$

Where ϕ_t^b is the price of a bond, and τ_t a lump-sum tax levied on all agents during the centralised trading session, both expressed in terms of fruit.

Asset Valuation - Lagos

The focus here is not on how the government should select the path $\{B_t, \tau_t\}$, but rather on characterising the equilibrium, and in particular asset prices and return, given such a path. Let $\mathbf{a}_t \in \mathbb{R}_+^2$ denote an agent's portfolio. Since they can now hold two assets, $\mathbf{a}_t = \left\{a_t^s, a_t^b\right\}$, where a_t^s and a_t^b denote the holdings of shares and bonds, respectively. Let ϕ_t^s be the price of s share in terms of fruit, and $\boldsymbol{\phi}_t = \left\{\phi_t^s, \phi_t^b\right\}$.

Asset Valuation - Lagos

The value function of an agent who enters the centralised market with portfolio $\mathbf{a}_t$ is a period when dividends are d_t now satisfies:

$$W\left(\mathbf{a}_t, d_t\right) = \max_{c_t, n_t, h_t, \mathbf{a}_{t+1}} \left\{U\left(c_t\right) + v\left(n_t\right) - Ah_t + \beta \mathbb{E} V\left(\mathbf{a}_{t+1}, d_{t+1}\right)\right\}$$

Subject to

$$c_t + w_t n_t + \boldsymbol{\phi}_t \mathbf{a}_{t+1} = \left(\phi_t^s + d_t\right) a_t^s + a_t^b + w_t h_t - \tau_t$$

$$0 \leq c_t, \quad 0 \leq n_t, \quad 0 \leq h_t \leq \bar{n}, \quad 0 \leq \mathbf{a}_{t+1}$$

Asset Valuation - Lagos

In the decentralise market, the terms at which a buyer with portfolio $\mathbf{a}$ trades with a seller with portfolio $\tilde{\mathbf{a}}$ are $[q(\mathbf{a}, \tilde{\mathbf{a}}), \mathbf{p}(\mathbf{a}, \tilde{\mathbf{a}})]$, where $q(\mathbf{a}, \tilde{\mathbf{a}}) \in \mathbb{R}_+$ is the quantity of special good traded, and $\mathbf{p}(\mathbf{a}, \tilde{\mathbf{a}}) \in \mathbb{R}_+^2$ represents the transfer of assets from the buyer to the seller.

The value of an agent who enters the decentralised market holding portfolio $\mathbf{a}$ in a period when the dividend realisation is x, satisfies

$$V(\mathbf{a}, x) = \alpha \int \{u[q(\mathbf{a}, \tilde{\mathbf{a}})] + W[\mathbf{a} - \mathbf{p}(\mathbf{a}, \tilde{\mathbf{a}}), x]\} \, d\mathbf{G}(\tilde{\mathbf{a}})$$
$$+ \alpha \int \{-e[q(\mathbf{a}, \tilde{\mathbf{a}})] + W[\mathbf{a} + \mathbf{p}(\mathbf{a}, \tilde{\mathbf{a}}), x]\} \, d\mathbf{G}(\tilde{\mathbf{a}}) + (1 - 2\alpha) W(\mathbf{a}, x)$$

Where $\mathbf{G}$ denotes the distribution of portfolios

Asset Valuation - Lagos

The terms of trade $[q(\mathbf{a}, \tilde{\mathbf{a}}), \mathbf{p}(\mathbf{a}, \tilde{\mathbf{a}})]$ are still determined by Nash bargaining where the buyer has all the bargaining power:

$$\max_{q_t, \mathbf{p}_t \leq \mathbf{a}_t} \{u(q_t) + W(\mathbf{a}_t - \mathbf{p}_t, d_t) - W(\mathbf{a}_t, d_t)\}$$

Subject to

$$-e(q_t) + W(\tilde{\mathbf{a}} + \mathbf{p}_t, d_t) \geq W(\tilde{\mathbf{a}}, d_t)$$

Let $\lambda_t^b = \frac{A}{w_t}$, $\lambda_t^s = \frac{A}{w_t}(\phi_t^s + d_t)$, and note that $W(\mathbf{a}_t + \mathbf{p}_t, d_t) - W(\mathbf{a}_t, d_t) = \boldsymbol{\lambda}_t \mathbf{p}_t$, where $\boldsymbol{\lambda}_t = \{\lambda_t^s, \lambda_t^b\}$

Asset Valuation - Lagos

These observations imply that the bargaining problem reduces to

$$\max_{q_t, \mathbf{p}_t \leq \mathbf{a}_t} \left\{ u\left(q_t\right) - \boldsymbol{\lambda}_t \mathbf{p}_t \right\}$$

Subject to

$$-e\left(q_t\right) + \boldsymbol{\lambda}_t \mathbf{p}_t \geq 0$$

The solution is $q\left(\boldsymbol{\lambda}_t \mathbf{a}_t\right)$, where the function $q\left(.\right)$ is still given by

$$q\left(\boldsymbol{\lambda}_t \mathbf{a}_t\right) = \begin{cases} q^* & \text{if } \boldsymbol{\lambda}_t \mathbf{a}_t \geq e\left(q^*\right) \\ e^{-1}\left(\boldsymbol{\lambda}_t \mathbf{a}_t\right) & \text{if } \boldsymbol{\lambda}_t \mathbf{a}_t < e\left(q^*\right) \end{cases}$$

Asset Valuation - Lagos

Using this bargaining solution and the linearity of $W(\mathbf{a}, d)$, the value of search can be written as:

$$V(\mathbf{a}, d) = \mathcal{S}(\boldsymbol{\lambda}\mathbf{a}) + W(\mathbf{a}, d)$$

In the centralise market, the agent's choices of c_t and n_t are still characterised by the same first order conditions as the first case. The first-order conditions for asset holdings, $\mathbf{a}_{t+1}$ are:

$$\frac{A}{w_t}\phi_t^i = \beta\mathbb{E}\frac{dV(\mathbf{a}_{t+1}, d_{t+1})}{da_{t+1}^i}, \quad \text{i = s, b}$$

Where $\frac{dV(\mathbf{a}_{t+1},d_{t+1})}{da_{t+1}^s} = \left[1 + \alpha\left(\frac{u'[q(\boldsymbol{\lambda}_{t+1}\mathbf{a}_{t+1})]}{e'[q(\boldsymbol{\lambda}_{t+1}\mathbf{a}_{t+1})]} - 1\right)\right]\lambda_{t+1}^s$ and $\frac{dV(\mathbf{a}_{t+1},d_{t+1})}{da_{t+1}^b} = \frac{dV(\mathbf{a}_{t+1},d_{t+1})}{da_{t+1}^s}\frac{\lambda_{t+1}^b}{\lambda_{t+1}^s}$. Again, the agent's choices do not depend on his individual asset holdings, and the distribution of assets will be degenerate in equilibrium.

Asset Valuation - Lagos

Given the dividend process $\{d_t\}_{t=0}^{\infty}$ and a path $\{B_t, \tau_t\}_{t=0}^{\infty}$, an equilibrium is an allocation $\{c_t, n_t, \mathbf{a}_{t+1}\}_{t=0}^{\infty}$, together with the set prices $\{w_t, \boldsymbol{\phi}_t\}_{t=0}^{\infty}$ and bilateral terms of trade $\{q_t\}_{t=0}^{\infty}$, such that:

(a) The individual choices $\{c_t, n_t, \mathbf{a}_{t+1}\}_{t=0}^{\infty}$ solve the agent's problem in the decentralised market, given prices;

(b) the terms of trade are determined by Nash bargaining, i.e., $q_t = \min\left\{q^*, e^{-1}\left[\frac{A}{w_t}\left(\phi_t^s + d_t\right)a_t^s + \frac{A}{w_t}a_t^b\right]\right\}$; and

(c) Prices are such that the centralised market clears, i.e., $c_t = d_t$, $a_{t+1}^s = 1$, and $a_{t+1}^b = B_{t+1}$

We will continue this model next lecture

References I

Ricardo Lagos. Asset prices and liquidity in an exchange economy. *Journal of Monetary Economics*, 57(8):913–930, 2010.